

Government Health Expenditure and Life Expectancy

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Abstract

Government healthcare expenditure is often associated with better population health outcomes, but the strength of this relationship may differ across countries and may reflect broader economic and regional differences. This study examines whether government healthcare expenditure is associated with life expectancy across countries from 2007 to 2017, and whether this relationship remains after accounting for GDP per capita, region, and year. Using 601 country-year observations from 64 countries, we estimate a simple linear regression model to examine the unadjusted relationship and a multiple linear regression model to examine the relationship after including the control variables. The simple model shows a positive and statistically significant association between government healthcare expenditure and life expectancy, supporting H1. However, the association becomes much smaller and is no longer statistically significant in the multiple regression model, so H2 is not supported. Overall, the findings suggest that healthcare expenditure alone does not fully explain cross-country differences in life expectancy and should be understood within a broader economic and national context.

Introduction

Health is a fundamental part of human well-being and an important concern for individuals, governments, and public policy. The World Health Organization defines health as “a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity” (Public Health Agency of Canada 2017). However, population health outcomes are shaped by a wide range of factors, including income, education, social and physical environments, and access to health services (Public Health Agency of Canada 2017). Understanding how public healthcare investment relates to population health is therefore important for evaluating health systems and informing public policy.

Government healthcare expenditure represents one way in which countries invest

public resources in population health. Greater public spending may improve access to healthcare services, disease prevention, treatment quality, and overall health system capacity, which may contribute to longer life expectancy. At the same time, countries differ substantially in their level of economic development, geographic context, and changes in health outcomes over time. These broader differences may influence both healthcare expenditure and life expectancy.

This project therefore examines the relationship between government healthcare expenditure and life expectancy across countries between 2007 and 2017. Life expectancy is used as a broad measure of population health because it reflects how long people are expected to live under current mortality conditions. Government healthcare expenditure is measured as a percentage of GDP, representing the share of national economic resources allocated by governments to healthcare.

Relevant Literature

Previous research generally suggests a positive association between healthcare expenditure and population health, although the strength of this relationship varies across settings. Studies of developing European Union countries have found a long-run relationship between public health expenditure and health outcomes. Greater public health expenditure is associated with higher life expectancy and lower infant mortality Onofrei et al. (2021). Similar evidence has been reported among OECD countries, where government health expenditure has been positively associated with life expectancy Anwar et al. (2023). Together, these findings suggest that greater investment in healthcare may contribute to better population health outcomes, supporting the expectation of a positive relationship between government healthcare expenditure and life expectancy.

However, the literature also shows that higher healthcare spending does not produce the same outcomes across countries. Research in the Middle East and North Africa has found that some countries spend relatively more on healthcare without achieving correspondingly higher life expectancy, suggesting that the allocation and efficiency of healthcare resources may also matter Balkhi et al. (2021). Meanwhile, a study of 175 countries also found that the relationship between health spending and life expectancy varies across countries. The benefits of additional health spending differ depending on a country's level of life expectancy Obrizan and Wehby (2018). These findings indicate that expenditure alone may not fully explain international differences in population health.

One important explanation for these differences is that life expectancy is influenced by broader economic and country-level conditions. Longitudinal research across OECD countries identifies not only healthcare expenditure, but also GDP, healthcare resources, social spending, demographic factors, and other contextual characteristics as important correlates of life expectancy Roffia et al. (2023). Anwar et al. also find that income, measured through GDP per capita, is positively associated with life expectancy alongside healthcare expenditure Anwar et

al. (2023). This suggests that an observed relationship between healthcare expenditure and life expectancy may partly reflect differences in national economic development rather than healthcare expenditure alone.

Therefore, to conclude, healthcare expenditure is generally associated with higher life expectancy, but this relationship is not uniform across countries and regions. Besides, economic and country differences mean that similar levels of healthcare spending are associated with different health outcomes. Much of the existing research focuses on particular groups of countries, such as developing EU countries, the MENA region, or OECD members, although some studies have examined broader international samples.

Building on this literature, our study examines government healthcare expenditure and life expectancy across 64 countries from 2007 to 2017. In addition to estimating the unadjusted relationship between the two variables, we extend the analysis by controlling for GDP per capita, region, and year. This allows us to examine whether the positive association between government healthcare expenditure and life expectancy remains after accounting for differences in economic development, geographic context, and changes over time.

Research Question and Hypotheses

This research aims to address two research questions by stating two related hypotheses.

RQ1: How is government health expenditure associated with life expectancy across countries from 2007 to 2017?

Based on the expectation that greater public healthcare spending may improve access to healthcare services, disease prevention, treatment quality, and overall health system capacity, we propose the following hypothesis:

H1: Countries with higher government healthcare expenditure as a percentage of GDP tend to have higher life expectancy.

This first research question examines the overall, unadjusted association between government healthcare expenditure and life expectancy.

RQ2: Does the relationship between government healthcare expenditure and life expectancy remain after accounting for GDP per capita, regional differences, and changes over time?

Because national income, geographic differences, and general changes over time may also be associated with population health, the final analysis incorporates GDP per capita, region, and year as control variables. We therefore propose a second hypothesis:

H2: The positive association between government healthcare expenditure and life expectancy will remain after accounting for GDP per capita, region, and

year.

Together, the two research questions allow us to compare the overall relationship between government healthcare expenditure and life expectancy with the relationship that remains after accounting for major country-level differences. Because the data are observational, the analysis focuses on associations rather than claiming a direct causal effect of government healthcare expenditure on life expectancy.

Dataset and Methods

Dataset

This study combines three international datasets to examine the relationship between government healthcare expenditure and life expectancy across countries between 2007 and 2017.

The main explanatory variable was obtained from the World Bank Data360 platform using the International Monetary Fund (IMF) Government Finance Statistics indicator IMF_COFOG_GEL_GF07 (Government Expenditure on Health), which measures general government healthcare expenditure as a percentage of GDP World Bank (2026b). The main outcome variable was obtained from the World Bank Data360 indicator WEF_GCIHH_LIFEEXPECT, which reports life expectancy measured in years World Bank (2026c). To account for differences in national economic development, GDP per capita was obtained from the World Bank World Development Indicators (WDI) using indicator WB_WDI_NY_GDP_PCAP_CD World Bank (2026a). A regional classification variable was additionally generated from ISO3 country codes using the countrycode R package.

The unit of analysis is the country-year. After cleaning and standardizing the three datasets, they were merged using ISO3 country codes and year to construct the final analytical dataset. The final dataset contains 601 country-year observations across 64 countries covering the period from 2007 to 2017.

One limitation of the dataset is that the historical Government Health Expenditure indicator (2007–2017) is no longer publicly available through the current World Bank Data360 or IMF Government Finance Statistics interfaces. In addition, because the final analysis requires complete information across all model variables, only countries with complete observations were retained in the analytical sample.

Data Preparation

Data acquisition, cleaning, and merging were performed using reproducible R scripts within an R Project workflow. Life expectancy and GDP per capita data were downloaded directly from the World Bank Data360 platform using

Table 1: Observations and countries retained at each stage of data preparation

Stage	Observations	Countries
Clean health expenditure data	673	75
Clean life expectancy data	1524	152
Clean GDP per capita data	2301	211
Health and life expectancy merged	601	64
Final complete-case analysis data	601	64

documented API requests and stored locally before analysis. The historical Government Health Expenditure dataset (IMF_COFOG_GEL_GF07) was originally downloaded during the project proposal stage. During preparation of the final project, attempts were made to reproduce the original download through both the current World Bank Data360 platform and the IMF Government Finance Statistics (GFS) Data Explorer. Because the historical 2007–2017 observations are no longer publicly accessible, the archived CSV retained from the proposal stage was used with instructor approval. The original data source, indicator information, retrieval process, and verification steps are documented in the project repository.

Each dataset was cleaned separately prior to merging. All datasets were restricted to the common study period of 2007–2017. For the health expenditure dataset, only general government expenditure observations measured as a percentage of GDP were retained. For the life expectancy dataset, observations reported in years were retained. GDP per capita observations were restricted to valid positive values, and the natural logarithm of GDP per capita was calculated because the original variable exhibited substantial right skewness. Country identifiers were standardized using ISO3 country codes, duplicate country-year observations were removed, and a regional classification variable was derived from ISO3 country codes using the countrycode package.

The cleaned datasets were merged using country code and year. A complete-case analytical dataset was then created by retaining only observations with complete information for all variables included in the regression analysis. The resulting dataset contained 601 country-year observations across 64 countries and was used for all subsequent exploratory analyses and regression models. Table 1 reports the number of observations and countries retained at each stage of this process.

The health expenditure dataset is the binding constraint on sample size. It contains the fewest countries of the three sources, so merging it with the life expectancy data reduces the sample to 601 country-year observations across 64 countries. Adding GDP per capita removed no further observations, meaning every country-year with expenditure and life expectancy data also had GDP data available.

Methods

Analytical Approach

The analysis proceeds in two stages that correspond to the two research questions. Both stages use linear regression estimated by ordinary least squares on the pooled country-year data.

The first stage estimates the unadjusted association between government health-care expenditure and life expectancy. This model addresses RQ1 and tests H1:

$$LifeExpectancy_{it} = \beta_0 + \beta_1 HealthExpenditure_{it} + \varepsilon_{it}$$

where i indexes countries and t indexes years. $LifeExpectancy_{it}$ is life expectancy in years, and $HealthExpenditure_{it}$ is government healthcare expenditure as a percentage of GDP. The coefficient β_1 represents the average difference in life expectancy associated with a one percentage point difference in healthcare expenditure, without accounting for any other country characteristics.

The second stage adds the three control variables. This model addresses RQ2 and tests H2:

$$LifeExpectancy_{it} = \beta_0 + \beta_1 HealthExpenditure_{it} + \beta_2 \log(GDPperCapita_{it}) + \sum_r \gamma_r Region_{ri} + \beta_3 Year_t + \varepsilon_{it}$$

Comparing β_1 across the two models shows how much of the unadjusted association remains once broader country-level differences are taken into account.

Control Variables

Three control variables were selected based on the patterns identified in the literature.

GDP per capita is included because national income is associated with both healthcare expenditure and life expectancy, and may therefore account for part of any observed relationship between them (Anwar et al. 2023; Roffia et al. 2023). The variable enters in natural logarithm form because raw GDP per capita is strongly right skewed. The logarithmic form also matches the common expectation that a given proportional increase in income is associated with a similar improvement in health outcomes at any income level, rather than a fixed absolute increase.

Region is included as a categorical variable to account for geographic and institutional differences that are not captured by income alone, such as differences in health system organization, disease environment, and historical context.

Europe and Central Asia serves as the reference category because it contains the largest number of observations in the sample. Each regional coefficient therefore represents the average difference in life expectancy relative to that group.

Year is included to account for general improvements in life expectancy over the study period that are common to all countries, such as advances in medical technology. Year is centred on the midpoint of the study period so that the intercept refers to a year within the observed range rather than to year zero.

Interpretation

The data are observational and the models describe associations rather than causal effects. Countries were not assigned levels of healthcare expenditure, and unmeasured characteristics may be related to both expenditure and life expectancy. A change in the healthcare expenditure coefficient between the two models therefore indicates that the unadjusted association overlaps with the control variables, not that healthcare expenditure causes or fails to cause changes in life expectancy.

Software

All data acquisition, cleaning, merging, exploratory analysis, and statistical modelling were conducted in R R Core Team (2024). Data manipulation and visualization were performed using the tidyverse collection of packages Wickham et al. (2019), including readr Wickham et al. (2024), dplyr Wickham et al. (2026), and tibble Müller and Wickham (2023). Reproducible file paths were managed using here Müller (2025). Initial data inspection and cleaning used janitor Firke (2024). API-based data retrieval was implemented using httr2 Wickham (2025), regional classifications were generated using countrycode Arel-Bundock et al. (2018), and regression model outputs were summarized using broom Robinson et al. (2026).

Results and Findings

Descriptive Statistics

Table 2 reports descriptive statistics for the four continuous variables in the analytical sample of 601 country-year observations.

Life expectancy averages 76.74 years and ranges from 51.00 to 84.28 years. Government healthcare expenditure averages 5.10 percent of GDP and ranges from 0.36 to 9.30 percent. GDP per capita is strongly right skewed: the mean of 29,355.63 US dollars exceeds the median of 22,405.64, and the maximum is more than one hundred times the minimum. Taking the natural logarithm produces a much more symmetric variable with a mean of 9.83 and a standard deviation of 1.08, which supports the decision to use the logarithmic form in the regression models.

Table 2: Descriptive statistics for the analytical sample (601 country-year observations, 64 countries)

Variable	N	Mean	SD	Min	Median	Max
GDP per capita (current US\$)	601	29,355.63	24,756.37	1,022.62	22,405.64	123,678.71
Health expenditure (% of GDP)	601	5.10	2.28	0.36	5.20	9.30
Life expectancy (years)	601	76.74	5.52	51.00	78.74	84.28
Log GDP per capita	601	9.83	1.08	6.93	10.02	11.73

Figure 1 shows the distribution of government healthcare expenditure across the sample.

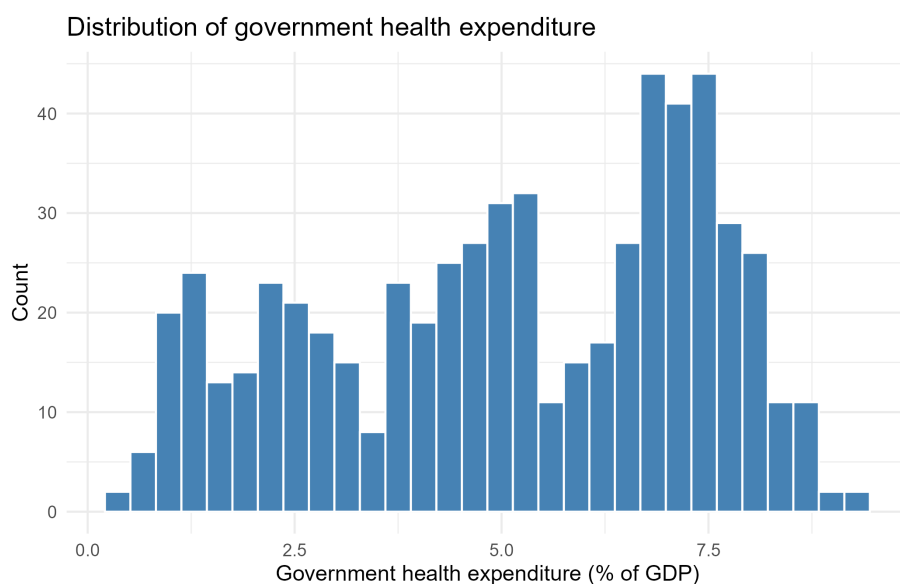


Figure 1: Distribution of government healthcare expenditure as a percentage of GDP

The distribution is spread across the full range rather than concentrated around a single value, with a larger group of observations between roughly 6.5 and 8 percent of GDP and a second group below 3 percent. This spread reflects substantial variation in how much countries allocate to public healthcare and provides useful variation for the regression analysis.

The Unadjusted Relationship

Figure 2 plots life expectancy against government healthcare expenditure with a fitted regression line.

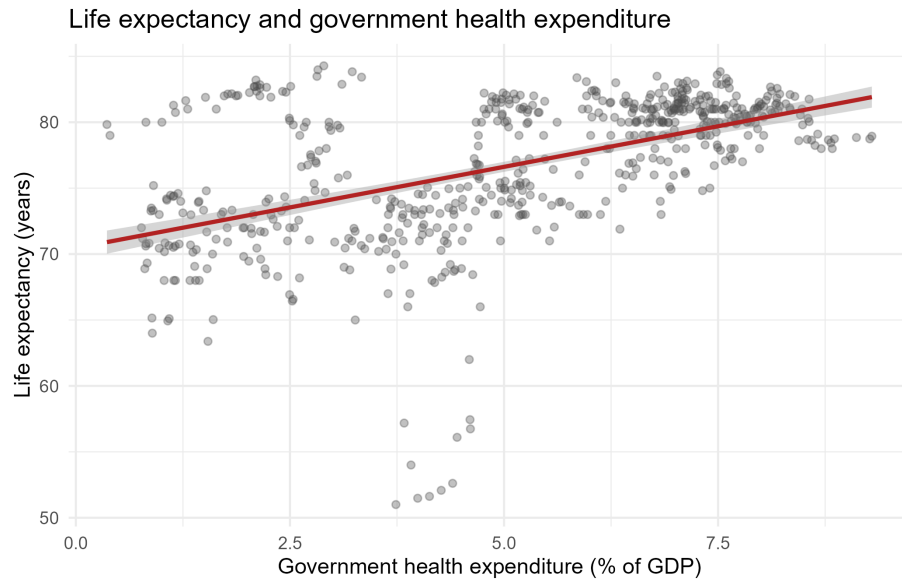


Figure 2: Life expectancy and government healthcare expenditure with fitted regression line

The fitted line slopes upward, indicating that countries with higher government healthcare expenditure tend to have higher life expectancy. However, the observations are widely scattered around the line. A distinct cluster of observations with life expectancy below 60 years appears at expenditure levels between roughly 3.5 and 5 percent of GDP, well within the middle of the expenditure range. This shows that similar levels of healthcare spending are associated with very different health outcomes, consistent with the literature reviewed earlier.

Figure 3 shows how life expectancy varies across the regional groups in the sample.

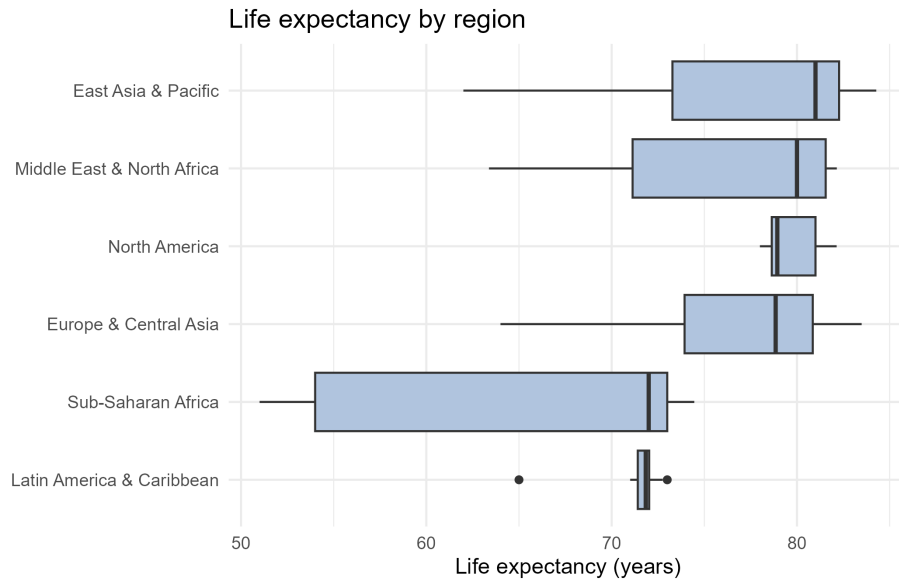


Figure 3: Life expectancy by region

Median life expectancy is broadly similar across East Asia and Pacific, Europe and Central Asia, the Middle East and North Africa, and North America, all sitting near or above 78 years. Sub-Saharan Africa shows a much wider spread extending well below 60 years. This pattern suggests that regional differences account for a meaningful share of the variation in life expectancy and supports including region as a control variable.

Regression Results

Table 3 reports the coefficients from both models.

In Model 1, the coefficient on government healthcare expenditure is 1.231 with a standard error of 0.085. The association is positive and statistically significant at any conventional level, with a 95 percent confidence interval of 1.064 to 1.399. A one percentage point higher share of GDP devoted to public healthcare is associated with approximately 1.23 additional years of life expectancy. Model 1 explains about 25.9 percent of the variation in life expectancy.

In Model 2, the same coefficient falls to 0.092 with a standard error of 0.071. The confidence interval of -0.047 to 0.231 includes zero, and the association is not statistically significant at the five percent level. The estimate is roughly one thirteenth of its unadjusted size, and the confidence interval no longer overlaps with the Model 1 interval. Adding the control variables raises the explained variation to about 75.8 percent.

Table 3: Regression results for Model 1 (unadjusted) and Model 2 (with controls)

Model	Term	Estimate	SE	p	95% CI
Model 1	(Intercept)	70.460	0.475	<0.001	[69.53, 71.39]
Model 1	Health expenditure	1.231	0.085	<0.001	[1.06, 1.40]
Model 2	(Intercept)	37.219	1.224	<0.001	[34.81, 39.62]
Model 2	Health expenditure	0.092	0.071	0.193	[-0.05, 0.23]
Model 2	Log GDP per capita	3.854	0.140	<0.001	[3.58, 4.13]
Model 2	East Asia & Pacific	1.452	0.344	<0.001	[0.78, 2.13]
Model 2	Latin America & Caribbean	1.416	0.726	0.052	[-0.01, 2.84]
Model 2	Middle East & North Africa	2.191	0.503	<0.001	[1.20, 3.18]
Model 2	North America	-1.049	0.633	0.098	[-2.29, 0.19]
Model 2	Sub-Saharan Africa	-9.177	0.627	<0.001	[-10.41, -7.95]
Model 2	Year (centred)	0.224	0.036	<0.001	[0.15, 0.29]

Among the control variables, log GDP per capita has the largest association with life expectancy. Its coefficient of 3.854 is highly significant, indicating that countries with higher national income tend to have substantially higher life expectancy. The regional coefficients are measured relative to Europe and Central Asia. The Sub-Saharan Africa coefficient of -9.177 is the largest in magnitude, indicating that these observations have markedly lower life expectancy even after accounting for income. The Middle East and North Africa and East Asia and Pacific coefficients are positive and significant but much smaller. The centred year coefficient of 0.224 indicates a general improvement of roughly one fifth of a year in life expectancy annually across the sample.

A formal comparison of the two models gives an F statistic of 174.12 on 7 and 592 degrees of freedom with a p-value below 0.001, confirming that the control variables jointly add substantial explanatory power.

Which Control Explains the Change

To identify which control variable is responsible for the change in the healthcare expenditure coefficient, each control was added to Model 1 separately.

Specification	Health expenditure coefficient
Model 1 (no controls)	1.232
Adding year only	1.236
Adding region only	1.267
Adding log GDP per capita only	0.032
Model 2 (all controls)	0.092

Adding year or region alone leaves the coefficient essentially unchanged. Adding

log GDP per capita alone reduces it to 0.032, close to zero. The change between the two models is therefore driven almost entirely by national income rather than by regional or time differences.

This result is consistent with the correlation of 0.634 between government healthcare expenditure and log GDP per capita in the sample. Wealthier countries tend both to spend a larger share of GDP on public healthcare and to have higher life expectancy. The unadjusted association in Model 1 therefore largely reflects differences in national income that are shared by both variables.

Model Diagnostics and Robustness

Figure 4 plots the residuals from Model 2 against the fitted values.

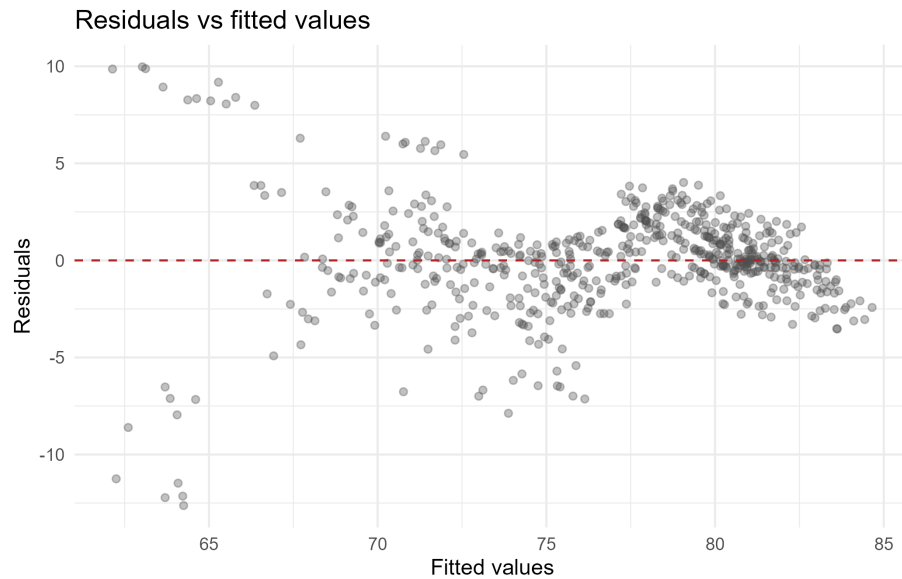


Figure 4: Residuals against fitted values for Model 2

Residuals are centred around zero across most of the fitted range, which supports the linear specification. However, the spread of residuals is noticeably wider at lower fitted values, with several residuals beyond plus or minus 8 years below a fitted value of 70. This indicates that the model predicts life expectancy less precisely for countries at the lower end of the distribution, and the standard errors for those observations should be interpreted with some caution.

An influence check identified 26 observations with a Cook's distance above the conventional $4/n$ threshold. The most influential observations are concentrated in a single country, South Africa between 2008 and 2013, which combines moderate healthcare expenditure of roughly 4 percent of GDP with life expectancy near 51

years. Re-estimating Model 2 without these influential observations raises the healthcare expenditure coefficient to 0.263. This is larger than the full-sample estimate but remains far below the unadjusted estimate of 1.231, so the main conclusion does not depend on these observations.

Hypothesis Assessment

H1 is supported. Countries with higher government healthcare expenditure as a percentage of GDP have higher life expectancy in the unadjusted analysis. The coefficient of 1.231 is positive, statistically significant, and consistent with the positive relationship reported in the existing literature.

H2 is not supported. The positive association does not remain after accounting for GDP per capita, region, and year. The coefficient falls from 1.231 to 0.092 and is no longer statistically distinguishable from zero. The sequential analysis shows that national income alone accounts for almost all of this change, indicating that the unadjusted relationship largely reflects differences in economic development rather than an independent association between public healthcare spending and life expectancy in this sample.

Discussion and Conclusion

The findings suggest that government healthcare expenditure should not be considered alone when comparing life expectancy across countries. Although countries that spend more on public healthcare tend to have higher life expectancy in the unadjusted analysis, this relationship becomes much weaker after accounting for broader country-level differences. This does not mean that healthcare spending is unimportant. Instead, it suggests that population health depends on a wider economic and social context, and that similar levels of healthcare spending may be associated with different outcomes across countries.

The research limitations come from the scope of the research question itself. First, life expectancy is a broad measure of population health. It summarizes mortality conditions but does not capture other important outcomes such as quality of life, disability, access to care, or differences in health across groups within the same country. Second, healthcare spending may affect health outcomes over many years. Spending in one year may therefore not be immediately reflected in life expectancy in that same year. In addition, countries organize and use healthcare resources differently. Two countries with similar spending levels may invest in different services, populations, or types of care, which may lead to different health outcomes.

The study also focuses on country-level averages. These national measures can hide important differences within countries, such as differences by income, age, gender, or location. In addition, the study covers only 2007 to 2017, so the findings describe a specific period and may not fully reflect more recent healthcare systems, technologies, or public health challenges.

Future research could examine more detailed health outcomes in addition to life expectancy, consider possible time lags between healthcare investment and health outcomes, and study how healthcare spending is allocated across different services. More recent data and subgroup-level data could also help explain why the same level of healthcare investment may produce different outcomes across countries and populations.

Overall, our results suggest that the important question is not simply whether countries spend more on healthcare, but also how, where, and under what conditions those resources are used. Healthcare expenditure is one component of population health, but it should be understood in the broader context in which healthcare systems operate.

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